HW 3

1. 60%

a. $P = 100\ lbf/in^2$ $u = 500\ btu$

$u_1 = u_f + x_1 u_{fg}$

$500 = 298.19 + x_1(1110.5 - 298.19)$

$\boxed{x_1 = .25}$

$h_1 = h_f + x_1 h_{fg}$

$= 298.51 + .25(1187.5 - 298.5)$

$= \boxed{520.75\ BTU/lb}$

$v_1 = v_f + x_1 v_{fg}$

$= .01774 + .25(4.4327 - .01774)$

$= 1.121\ ft^3/lb$

$\boxed{T = 327.8\,F}$

6. $v_2 = v_f + x_2 v_{fg}$

$.02533 + .5 (3.4664 - .02533)$

$\boxed{2 \ ft^3/lb}$

$v_2 = v_f + x_2 v_{fg}$

$85.91 + .5 (662.42 - 85.93)$

$\boxed{327.2 \ Btu/lb}$

$p_2 = 73.359 \ lbf/in^2$

c. $p = 30698$ $\quad V_3 = 0.24834 \ m^3$

$$\boxed{\begin{aligned} T &= 430°C \\ u &= 3022.2 \ kJ/kg \\ h &= 3342.4 \ kJ/kg \end{aligned}}$$

4. $p = 1.3299 \ bar$

$v_4 = .1464 \ m^3/kg$

$V_4 = 216.34$

2. $T_I = 180\,°C$

$T_f = 100\,°C$

A. $\boxed{1.002\ MPa}$

B. $v_2 = v_1$

v

$v =$

$v_g = .1941\ m^3/kg$

$v_2 = v_f + x_2(v_g - v_f)$

$.1941 = .$

$100\,°C$

$v_f = .00104346\ m^3/kg$

$v_g = 1.6718\ m^3/kg$

$.1941 = .00104346 + x_2(1.6718 - .00104346)$

$1.67075654\,x_2$

$\frac{.019306}{1.67075654}$

$\boxed{x_2 = .115 = 11.5\%}$

3. $m = 2\ kg$

$P = 300\ kPa = 3\ bar$

$V_f = 2.064\ m^3$

$v_f = 1.0732 \times 10^3\ m^3$

$v_g = 0.6058\ m^3/kg$

$v_1 = 0.6058 \times 2$

A. $\boxed{133.6°C}$

B. State 2 = $V_2 = V_1$

P 3 bar

Specific volume $\frac{V_2}{m} = \frac{2.064\ m^3}{2}$

$1.032\ m^3/kg$

$\boxed{408°C}$

C. $W = P(V_2 - V_1)$

$300\ kPa \times (2.064 - 1.2116)$

$\boxed{255.72\ [kJ]}$

F

D.

$P1 = 10\,bar$ $\quad$ $P_2 = 3\,bar$

$T1 = 60°$ $\quad$ $T_2 = 30°C$

$W = 20\,kJ$ $\quad$ $m = 1\,kg$

A. $V1 = m\,v_1$ $\quad$ $P1 = 10\,bar$

$V1 = 1(v1)$ $\quad$ $T = 60°$

$V1 = 1(.02301)$

$V1 = \boxed{.02301\,m^3}$

b. $Q = W + m(u_2 - u_1)$ $\quad$ $u1 = 263.35$

$20\,kJ + 1($ $\quad$ $u2 = 253.061$

$Q = 20($

$20 + ($

$Q = \boxed{4.711\,kJ}$

8. $m = 1.50 kg$ $T_1 = 20°$

$T_2 = 120°$

9.

9. $P = \boxed{31.16 \text{ kPa}}$

6. $T_1 = 20°$ $v_f = .001023 \text{ m}^3/\text{kg}$

$v_g = 5.0463 \text{ m}^3/\text{kg}$

$T_2 =$ $v_g = .39163 \text{ m}^3/\text{kg}$

$m = .11$

$Q = m(h_2 - h_1)$ $h_{fg} = \cancel{2706}$

$h_1 = 689.78 \text{ kJ/kg}$

$h_2 = 2706.0$

$Q = \boxed{3024.33 \text{ kJ}}$

808

B.. P 10 bar

$m = 25kg$

$a = 60\%$

$v_f = .33352 \times 10^{-3}$ $v_g = .0236 \frac{m^3}{kg}$

A.

$v = .01449\ m^3kg$

$x = \frac{m_g}{m_{37.25}}$ $m_g = 37..5$

$V = .9036\ m^3$

b. $v_g(m - m_f)$

$= .85\ m^3$

$= \frac{.833}{.9036} = .977 = 97.7\%$

b)

a) $m = .2\,g$

$T_i = 27°C = 300K$

$T_f = 17°C = 290K$

$P_2 = .30\ bar$

$P_1 = 1.3\ bar$

$$\frac{R}{m} \quad \frac{8.314}{4} = 2.0785\ \frac{KJ}{Kg \cdot K}$$

$$V1 = \frac{mR_{He}\,T1}{P1}$$

$= \cancel{9.59 m}$ $\boxed{.959 \times 10^{-3}\ m^3}$

$$V2 = \frac{mR_{He}\,T2}{P2} = \boxed{1.508 \times 10^{-3}\ m^3}$$

8. $T_1 = 25\,°C = 298.15\,°K$

$V_1 = .05\,m^3$

$T_f = 257\,°C = 530.15\,K$

A. $Q = 2.75\,KJ$ X

$P_1V_1\ mRT$

$m = \frac{P_1V_1}{RT}$

$2.75 \cdot .08 = m$

$m \cdot RT$

60%

$\frac{293.18\,K}{8.3114\,T}$

$\boxed{m = .00084837\,kg}$ X